Showing our fibre network on a web map: 3 options compared

We need a web map that shows our fibre sites and the cable routes between them. This page compares three ways to build it, using the same mock data (5 sites and 4 routes around Kajang, Cheras, KL, Bangi and Putrajaya).

On every map, routes are coloured by type: ▬ Underground (brown, solid) and ▬ Aerial (blue, dashed). Click any site or route to see its name and details.

Printed copy: the live maps are left out. See the Screenshots section.

Screenshots

Captured with Playwright (headless Chromium) at 1400 × 800 on 15 Oct 2026, so this record still works when the servers are off. Click an image for full size.

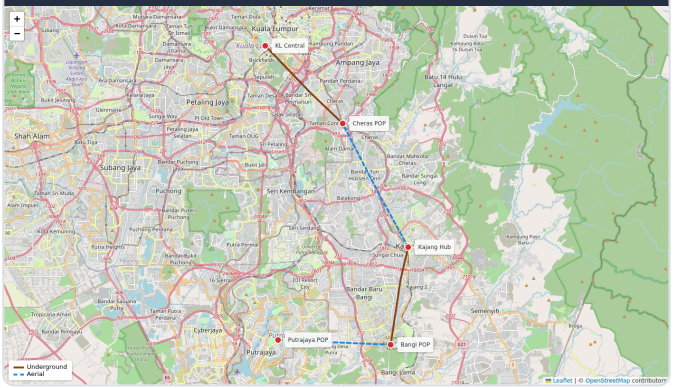
Whole network

Google Maps

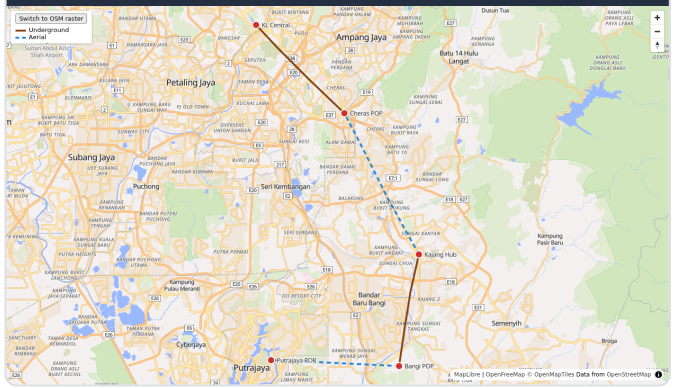
Not tested

Needs a Google API key and a billing account with a card.

Leaflet + OpenStreetMap



MapLibre + OpenFreeMap



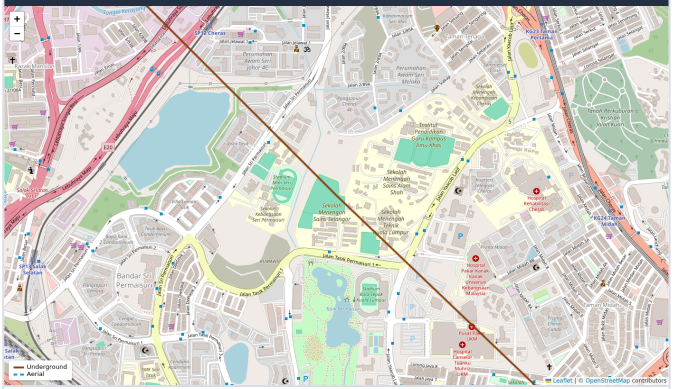
Close-up on one route (Cheras to KL Central, same scale)

Google Maps

Not tested

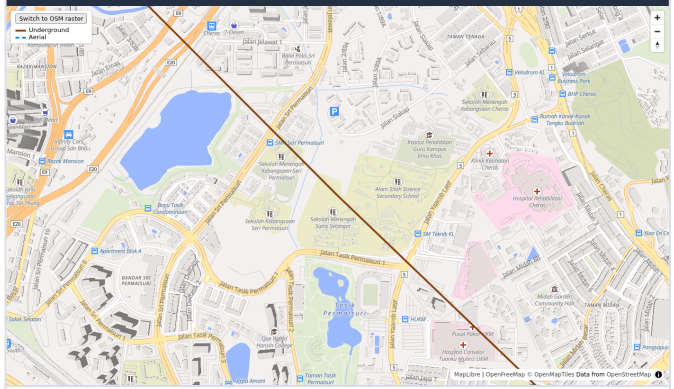
Needs a Google API key and a billing account with a card.

Leaflet + OpenStreetMap



Raster images: text and lines are fixed pixels and blur when scaled between zoom levels.

MapLibre + OpenFreeMap



Vector data drawn by the browser: text and lines stay sharp at any zoom.

Comparison

	Google Maps	Leaflet + OSM tiles	MapLibre + OpenFreeMap
Look	Most polished and familiar basemap	Flat raster tiles, looks dated, blurs between zoom levels	Smooth vector rendering, crisp at any zoom, rotation and tilt
Route styling	Data layer only: colour and width, no dashes	Full control, dashes included	Full control, data-driven styling, dashes via separate layers
Cost	Free monthly allowance, then pay per use	Free	Free
Key or account needed	API key plus billing account with a card	None	None
Integration effort	Script tag plus key, loadGeoJson	Easiest, about 20 lines	Slightly more: style, sources and layers

	Google Maps	Leaflet + OSM tiles	MapLibre + OpenFreeMap
Scales to many routes	Fine for hundreds	Fine for hundreds to low thousands, slows after that	Best: WebGL handles tens of thousands
Risks	Vendor lock-in, key management, terms of use	OSM's public tile server forbids heavy use	OpenFreeMap has no SLA unless you self-host
Self-hosting/offline	No	Yes, with your own tiles	Yes, with your own tiles and style
PostGIS path	GeoJSON	GeoJSON	GeoJSON now, vector tiles later (Martin or pg_tileserv)

Integration

The code that loads the route data and draws the routes, copied from each page. All three read the same `routes.geojson` file exported from our PostGIS database.

Google Maps not tested

```
map.data.setStyle(f => {
  // ... site marker style skipped ...
  return { strokeColor: isUnderground(desc) ? '#8B4513' : '#1E88E5', strokeWeight: 4 };
});
map.data.loadGeoJson('routes.geojson', null, done);
```

google.html, lines 29, 37, 38 and 48

Leaflet + OpenStreetMap

```
Promise.all(['routes.geojson', 'sites.geojson'].map(f => fetch(f).then(r => r.json()))).then(([routes, sites]) => {
  const r = L.geoJSON(routes, {
    style: f => isUnderground(f.properties.description)
      ? { color: '#8B4513', weight: 4 }
      : { color: '#1E88E5', weight: 4, dashArray: '8 6' },
```

osm.html, lines 34–38

MapLibre + OpenFreeMap

```
map.addSource('routes', { type: 'geojson', data: data.routes });
map.addLayer({ id: 'routes-underground', type: 'line', source: 'routes', filter: UNDERGROUND,
  layout: { 'line-cap': 'round' }, paint: { 'line-color': '#8B4513', 'line-width': 4 } });
map.addLayer({ id: 'routes-aerial', type: 'line', source: 'routes', filter: ['!', UNDERGROUND],
  paint: { 'line-color': '#1E88E5', 'line-width': 4, 'line-dasharray': [2, 1.5] } });
```

maplibre.html, lines 48 and 50–53

Recommendation for this project

Use **MapLibre with the OpenFreeMap basemap**. It costs nothing, needs no key or card, and gave the sharpest result in our test. It also has the most room to grow: it can show tens of thousands of routes, and later it can read map tiles straight from our PostGIS database (with Martin or pg_tileserv) instead of one large file. The main risk is that OpenFreeMap is a free service with no uptime guarantee. Before real users depend on it, we should host the basemap ourselves or move to a paid tile provider; neither change needs a rewrite. Leaflet is a fine fallback for a quick internal prototype, but its look and its public tile server limits make it a weaker long-term choice. Google Maps was not tested, because it needs a billing account with

> a card. It is only worth revisiting if a familiar Google look matters more than cost and vendor lock-in.